BRIAN JARRETT

Solutions Architecture & Technical Leadership | Systems Automation, Robotics Design & Subsystem Testing

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PROFESSIONAL SUMMARY

Hands-on Technical Leader and Solutions Architect with a 30-year systems engineering foundation spanning enterprise bare-metal compute, private cloud infrastructure, and federal compliance, alongside over a decade designing competitive robotic systems and authoring real-time Python software controlling swerve drive kinematics, articulated mechanisms, and vision targeting. A practical problem-solver who understands systems from the ground up, verifies data directly at the source, and approaches platforms with rigorous testing and automation discipline—from early career software QA leadership and enterprise cloud orchestration testing to Hardware-in-the-Loop (HIL) subsystem validation and tournament diagnostics. Proven track record of entering unfamiliar technical ecosystems, rapidly deconstructing complex architectures, and translating intricate enterprise and federal requirements (FedRAMP / STIGs) into reliable, automated platforms. Demonstrated leadership directing enterprise zero-day incident response across 6,000+ nodes to defend FedRAMP ATO with the HHS PMO, building custom vulnerability data platforms, and serving as the technical bridge across multidisciplinary software, networking, and platform teams.

SKILLS

Solutions & Data Engineering:
- Solutions Architecture & Prototyping
- Python (FastAPI, Pandas, SQLAlchemy)
- Robotics Design & Subsystem Testing
- Validation & QA Testing (HIL)
- ETL Pipelines & PostgreSQL

Systems & Distributed Scale:
- Cisco UCS & Bare-Metal Compute
- OpenStack (Nova, Heat)
- vSphere/ESXi, Proxmox/KVM, Xen
- Ceph Block Storage (RBD)
- HA Clustering & Linux Systems

DevOps, Security & Leadership:
- FedRAMP Moderate & DoD STIGs
- Systems & Platform Automation
- Qualys API & Vulnerability Mgmt
- FreeIPA/LDAP, RBAC/HBAC
- Technical Leadership & Mentoring

PROFESSIONAL EXPERIENCE

Cisco Systems | Cloud Engineering Technical Leader

Aug 2019 – July 2026 (Remote) * **Federal ATO Governance & Enterprise Log4j Response (HHS PMO):** Played an integral technical role in the enterprise Log4j response across 6,000+ systems for the HHS PMO (governing FedRAMP ATO). Built custom Ansible/Python inspection tooling locating nested libraries missed by commercial scanners, verified remediation, and compiled PMO evidence packages defending the ATO. * **Vulnerability Data Platform & Fleet UI Architecture (FastAPI, PostgreSQL, Tableau):** Designed and maintained an internal FastAPI + PostgreSQL vulnerability portal (fed by automated Jira ETL) used by engineering teams to triage issues and by leadership to track fleet status across the FedRAMP footprint. Extended platform with Tableau dashboards tracking SLA severity. * **Cross-Discipline Technical Translation & Federal Deviation Authority:** Served as technical bridge across software, storage, and networking teams, translating complex CVEs and DISA STIGs into actionable remediation paths. Authored formal technical compliance deviation and risk justification packages submitted to the federal PMO. * **Enterprise Modernization & Qualys Grouping Taxonomy (ServiceNow Migration):** Architected Qualys asset grouping and tagging taxonomy for enterprise ETL migration to ServiceNow, ensuring automated classification and routing of compliance findings across engineering teams. * **DevSecOps & Compliance Automation (STIG XCCDF/XML):** Engineered Python tools parsing raw STIG XCCDF (XML) data with the Qualys API to generate audit-ready checklists, eliminating hundreds of manual audit hours. * **Engineering Mentorship & Technical Guidance:** Mentored junior and mid-level engineers on Linux diagnostics, root-cause troubleshooting, and operational practices. Supported hiring through technical interviewing and structured onboarding across distributed remote teams. * **Skills:** Python (FastAPI, Pandas), PostgreSQL, REST APIs (Jira, Qualys), ETL Pipelines, Tableau, Ansible, RHEL/Ubuntu, Cisco UCS, STIG/FedRAMP Moderate, Federal ATO Governance.

Cisco Systems | Cloud Systems Engineer & Lead Compute Engineer

Jan 2017 – Aug 2019 (Remote) * **Enterprise Identity Architecture & SOC Collaboration (FreeIPA/HBAC):** Architected and deployed an automated, high-availability FreeIPA cluster in OpenStack via Ansible to replace legacy 389 DS. Partnered with the SOC to establish fine-grained Host-Based Access Control (HBAC) policies, guiding application teams on service permissions and standardizing access control across 2,000+ hosts. * **Lead Compute Engineering & Virtualization Operations:** Managed enterprise virtualization clusters on VMware vSphere and Cisco UCS bare metal, serving as lead compute engineer maintaining high availability, strict federal SLAs, and supporting Cisco MDS Fibre Channel SAN switch fabrics. * **Compute Security & Vulnerability Remediation:** Hardened VMware vSphere clusters, Cisco UCS bare metal, and FedRAMP bastion environments, driving continuous vulnerability remediation under strict federal uptime and security SLAs. * **Secure Access & Bastion Infrastructure (Bastions, LDAP, Active Directory):** Architected and maintained secure access infrastructure across Webex commercial and FedRAMP footprints, managing hardened Linux bastion hosts, Active Directory domain controllers, and LDAP directory services across 2,000+ systems. * **Skills:** Ansible,

FreeIPA (HA/HBAC), OpenStack, VMware vSphere, Cisco UCS, RHEL/CentOS, Windows Server, RBAC/HBAC, Qualys, FedRAMP Compliance.

Select Group @ Cisco | Cloud DevOps/QA Engineer (Contract)

Feb 2016 – Jan 2017 (Remote) * **Performance Optimization:** Identified and removed redundant **Nova Scheduler** filters in **OpenStack**, reducing instance scheduling time for 700 nodes from 20 seconds to 5 seconds (a **75% improvement** in orchestration speed). * **Bare-Metal PXE Provisioning & Automation at Scale:** Automated bare-metal server deployment and configuration via **PXE boot** and **Ansible** using lights-out management (LOM) interfaces across hundreds of production servers. * **Cloud QA & Orchestration:** Managed global test cycles for Compute and Heat components during cloud upgrades. Developed **Python**, **Bash**, and **Ansible** scripts to perform comprehensive automated orchestration tests. * **Skills:** OpenStack, Ansible, Python, Bash, RHUI, Proxmox, CentOS/RHEL/Windows, Performance Tuning, Cloud QA.

Australian Laboratory Services | Senior Systems & Storage Administrator

Sep 2014 – Jan 2016 (Grand Junction, CO) * **Distributed Storage & Virtualization:** Engineered an Ubuntu-based **Ceph cluster** for **VMware ESXi** hosts, benchmarking iSCSI vs. RBD performance. Used **Ceph RBD snapshots** for automated schema migration testing with zero errors. * **Infrastructure Automation:** Wrote custom **Python (SQLAlchemy)** verification scripts to audit SQL foreign keys during data migration. Implemented **SaltStack** to automate Linux VM and Ceph node provisioning. * **System Stability & Monitoring:** Integrated **Nagios** and **Observium** to monitor network operations, reducing service restarts. Diagnosed WAN/VPN bottlenecks and upgraded gateway hardware to double network throughput. * **Skills:** Ubuntu, Ceph RBD, VMware ESXi, SaltStack, Python, SQLAlchemy, Nagios, Observium, MySQL, Asterisk, VPN Routing.

Garfield County School District 16 | Senior Systems/Network Administrator

Oct 2002 – Jun 2014 (Parachute, CO) * **Systems Integration & Automation:** Programmed a custom **Flask (SQLAlchemy)** app syncing Student Info Systems with **Active Directory** and **Google Apps**, reducing account helpdesk tickets by **97%**. * **Enterprise Modernization & Development:** Replaced legacy databases with centralized PostgreSQL platforms, and automated state compliance reporting via custom PHP scripts. * **Infrastructure & Systems Virtualization:** Centralized desktop computing with **Windows Terminal Servers** and **HP Thin Clients** for 6 labs, saving **$60k+** in CAPEX. * **Network Engineering:** Designed a high-speed WAN using **Ruckus wireless bridges**, increasing bandwidth by **6,500%** (1.5Mbps to 100+ Mbps) with zero monthly service costs. Deployed district-wide Cisco CallManager. * **Skills:** Python, Flask, SQLAlchemy, PHP, PostgreSQL, XenServer, VirtualBox, Cisco Routers/Switches/PIX, Cisco CallManager, Ruckus Wireless.

Foundational Experience | Systems Engineering, Software QA & Avionics Foundations

1991 – 2002 (Various) * **Manager of QA Testing & Solutions Engineering (StorageSoft/Phoenix Tech):** Merged and managed QA testing with technical support for **ImageCast** disk cloning tools; hired and trained technicians, structured automated testing workflows across heterogeneous Windows/Linux labs, and acted as systems integration engineer for enterprise clients. * **Airborne Radar Journeyman (U.S. Air Force):** Repaired high-power microwave systems, high-voltage circuits, and analog/digital systems on AWACS aircraft. * **Skills:** Analog & Digital Diagnostics, PC Architecture, Disk Cloning (ImageCast), Software QA, Network Design, Technical Training.

EDUCATION & CERTIFICATIONS

- **Linux Academy / A Cloud Guru:** AWS & Red Hat Enterprise Linux Curriculum
- **Regis University:** C Programming & Unix Systems Courses
- **Lincoln Christian University:** Music Theory & Composition
- **Colorado School of Mines:** 2 Years – Electrical Engineering & Applied Physics
- **Community College of the Air Force:** Avionics/Electronics Technician – RADAR Specialist (AWACS)
- **On the Job Training – Tinker AFB, OK:** AWACS RADAR and IFF Systems Journeyman – Awarded Senior Airman Below-the-Zone
- **Air Force Technical Training School – Keesler AFB, MS:** Electronics and AWACS RADAR Courses – Honor Graduate (99% Academic Test Average)

For a list of legacy certifications, see linkedin.com/in/bjarrett/details/certifications/